

THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

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ABSTRACT. We give a classical proof of the generalization of the uniform $\mathbb{G}_0$ dichotomy to families of digraphs indexed by elements of $\kappa^{\mathbb{N}}$. Using this, we obtain a classical proof of Kanovei's generalization of the $\mathbb{G}_0$ dichotomy to κ -Souslin digraphs.

INTRODUCTION

Suppose that κ is an infinite cardinal and X is a Hausdorff space. Endow κ with the discrete topology and $\kappa^{\mathbb{N}}$ with the product topology. For all $n \in \mathbb{N}$ and $s \in \kappa^n$, set $|s| = n$. For all $\sigma \in \kappa^{\leq \mathbb{N}}$, let $s \sqsubseteq \sigma$ indicate that $\forall i < |s| \ s(i) = \sigma(i)$. Define $\mathcal{N}_s = \{\sigma \in \kappa^{\mathbb{N}} \mid s \sqsubseteq \sigma\}$. We write $\forall^* \sigma \in \kappa^{\mathbb{N}} \ \phi(\sigma)$ to indicate that $\{\sigma \in \kappa^{\mathbb{N}} \mid \phi(\sigma)\}$ is comeager. The *projection* function from $\prod_{i \in I} X_i$ to X_i is given by $\text{proj}_{X_i}(x) = x(i)$. The *horizontal and vertical sections* of a set $R \subseteq X \times Y$ are given by $R^y = \{x \in X \mid x R y\}$ and $R_x = \{y \in Y \mid x R y\}$. Given a function $\pi: R \subseteq X \times Y \rightarrow Z$, define $\pi_x: R_x \rightarrow Z$ by $\pi_x(y) = \pi(x, y)$.

We say that $\phi: C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \rightarrow \kappa^{\mathbb{N}} \times X$ is *parameter preserving* if $\text{proj}_{\kappa^{\mathbb{N}}} \circ \phi = \text{proj}_{\kappa^{\mathbb{N}}} \upharpoonright C$, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly analytic (vertical) sections* if there are a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$.

The *diagonal union* of a sequence $(X_s)_{s \in \kappa^{< \mathbb{N}}}$ of subsets of $\kappa^{\mathbb{N}} \times X$ is given by $\bigvee_{s \in \kappa^{< \mathbb{N}}} X_s = \bigcup_{s \in \kappa^{< \mathbb{N}}} X_s \cap (\mathcal{N}_s \times X)$. Countable unions are expressible as diagonal unions, and the vertical sections of diagonal unions are themselves countable unions. We say that a set $B \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly Borel (vertical) sections* if it is in the smallest family of subsets of $\kappa^{\mathbb{N}} \times X$ containing the products of $\kappa^{\mathbb{N}}$ with open subsets of X and closed under complements and diagonal unions. We say that a function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ has *fibers with uniformly Borel (vertical) sections* if $c^{-1}(\{n\})$ has uniformly Borel sections for all $n \in \mathbb{N}$.

A digraph on X is an irreflexive set $G \subseteq X \times X$. A *homomorphism* from a digraph G on X to a digraph H on Y is a function $\pi: X \rightarrow Y$ such that $(\pi \times \pi)(G) \subseteq H$. A *Y -coloring* of G is a homomorphism $c: X \rightarrow Y$ from G to inequality. Fix $\mathfrak{s} \in \prod_{n \in \mathbb{N}} 2^n$ such that $\mathfrak{s}(\mathbb{N})$ is dense in $2^{< \mathbb{N}}$ and define $\mathbb{G}_0 = \bigcup_{n \in \mathbb{N}} \{(\mathfrak{s}_n \frown (i) \frown c)_{i < 2} \mid c \in 2^{\mathbb{N}}\}$.

The uniform $\mathbb{G}_0$ dichotomy for digraphs indexed by $\mathbb{N}^{\mathbb{N}}$ is a straightforward corollary of the $\mathbb{G}_0$ dichotomy itself (see [Mil12, Theorem 31]). Here we show that the following generalization to digraphs indexed by $\kappa^{\mathbb{N}}$ is a consequence of the classical proof of the $\mathbb{G}_0$ dichotomy:

Theorem 1. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ has uniformly analytic sections that are digraphs on X . Then exactly one of the following holds:*

- (1) *There is a function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$, whose fibers have uniformly Borel sections, such that c_σ is a coloring of G_σ for all $\sigma \in \kappa^{\mathbb{N}}$.*
- (2) *There is a continuous homomorphism $\pi: 2^{\mathbb{N}} \rightarrow X$ from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^{\mathbb{N}}$.*

In §1, we note closure properties of sets with uniformly analytic and Borel sections. In §2, we prove a Baire category fact about such sets. In §3, we establish a separation theorem for such sets. In §4, we show Theorem 1. And in §5, we obtain Kanovei's κ -Souslin generalization [Kan97, Abstract Theorem 1] of the $\mathbb{G}_0$ dichotomy.

We leave it to the reader to check that these results can be combined with the arguments establishing the usual consequences of the $\mathbb{G}_0$ dichotomy to obtain analogous uniform and κ -Souslin generalizations.

1. POINTCLASSES

We direct the reader to [Kec95, §14] for the most basic facts concerning analytic sets. The following closure properties can be proven in the same fashion as the corresponding facts for analytic sets:

Proposition 1.1. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under countable intersections and countable unions.*

Proposition 1.2. *Suppose that κ is an aleph, X and Y are Hausdorff spaces, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $B \subseteq \kappa^{\mathbb{N}} \times Y$ have uniformly analytic sections, and $f: A \rightarrow \kappa^{\mathbb{N}} \times Y$ is continuous and parameter preserving. Then $f(A)$ and $f^{-1}(B)$ have uniformly analytic sections.*

Given a set $S \subseteq \kappa^{<\mathbb{N}}$ and a sequence $(X_s)_{s \in S}$ of subsets of $\kappa^{\mathbb{N}} \times X$, define $\bigvee_{s \in S} X_s = \bigvee_{s \in \kappa^{<\mathbb{N}}} X_s$, where $X_s = \emptyset$ for all $s \in \sim S$.

Proposition 1.3. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under diagonal unions.*

Proof. Suppose that $(A_s)_{s \in \kappa^{<\mathbb{N}}}$ is a sequence of subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections and $A = \bigvee_{s \in \kappa^{<\mathbb{N}}} A_s$. Then there are closed

sets $C_s \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_s: C_s \rightarrow \kappa^{\mathbb{N}} \times X$ with $A_s = \phi_s(C_s)$ for all $s \in \kappa^{<\mathbb{N}}$. As the set $C_n = \bigcap_{s \in \kappa^n} C_s$ is closed and the function $\phi_n = \bigcup_{s \in \kappa^n} \phi_s \upharpoonright (C_s \cap (\mathcal{N}_s \times \mathbb{N}^{\mathbb{N}}))$ is continuous, the set $A_n = \bigcap_{s \in \kappa^n} A_s = \phi_n(C_n)$ has uniformly analytic sections for all $n \in \mathbb{N}$, as does $A = \bigcup_{n \in \mathbb{N}} A_n$ by Proposition 1.1. $\square$

Proposition 1.4. *Suppose that κ is an aleph and X is a topological space. Then every Borel subset of $\kappa^{\mathbb{N}} \times X$ has uniformly Borel sections.*

Proof. It is sufficient to show that every open set $U \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. For all $s \in \kappa^{<\mathbb{N}}$, let $U_s \subseteq X$ be the maximal open set for which $\mathcal{N}_s \times U_s \subseteq U$. Then $U = \bigcap_{s \in \kappa^{<\mathbb{N}}} \kappa^{\mathbb{N}} \times U_s$. $\square$

Proposition 1.5. *Suppose that κ is an aleph, X and Y are topological spaces, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $B \subseteq \kappa^{\mathbb{N}} \times Y$ have uniformly Borel sections, and $f: A \rightarrow \kappa^{\mathbb{N}} \times Y$ is Borel and parameter preserving. Then $f^{-1}(B)$ has uniformly Borel sections.*

Proof. Proposition 1.4 ensures that the family of sets for which the proposition holds contains the open sets, and it is clearly closed under complements and diagonal unions (since f is parameter preserving). $\square$

Proposition 1.6. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly analytic sections, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then $A \cap B$ has uniformly analytic sections.*

Proof. By the usual proof of the corresponding fact for analytic and Borel sets, we need only note that if $B = \bigcap_{s \in \kappa^{<\mathbb{N}}} B_s$ and both $A \cap B_s$ and $A \setminus B_s$ have uniformly analytic sections for all $s \in \kappa^{<\mathbb{N}}$, then $A \cap B = \bigcap_{s \in \kappa^{<\mathbb{N}}} A \cap B_s$ has uniformly analytic sections by Proposition 1.3, and $A \setminus B = \bigcap_{n \in \mathbb{N}} \bigcap_{s \in \kappa^n} A \setminus B_s$ has uniformly analytic sections by Propositions 1.1 and 1.3. $\square$

2. BAIRE CATEGORY

We direct the reader to [Kec95, §8] for the basic facts concerning Baire category used below. Although $\kappa^{\mathbb{N}}$ is not separable when κ is uncountable, localization holds in every completely metrizable space.

Proposition 2.1. *Suppose that κ is an aleph, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then every horizontal section of B has the Baire property.*

Proof. It is enough to show that if $n \in \mathbb{N}$ and the horizontal sections of $B_s \subseteq \kappa^{\mathbb{N}} \times X$ have the Baire property for all $s \in \kappa^{<\mathbb{N}}$, then the horizontal sections of $B = \bigcap_{s \in \kappa^n} B_s$ also have the Baire property. Given $x \in X$, fix an open set $U_s \subseteq \kappa^{\mathbb{N}}$ and dense open sets $U_{m,s} \subseteq \kappa^{\mathbb{N}}$ such that

$(B_s^x \triangle U_s) \cap \bigcap_{m \in \mathbb{N}} U_{m,s} = \emptyset$ for all $s \in \kappa^n$. Then $U = \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap U_s$ is open, $U_m = \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap U_{m,s}$ is dense and open for all $m \in \mathbb{N}$, and $(B^x \triangle U) \cap \bigcap_{m \in \mathbb{N}} U_m = \emptyset$, so B^x has the Baire property. $\square$

3. SEPARATION

Recall that $B \subseteq X$ separates $A_0, A_1 \subseteq X$ if $A_0 \subseteq B$ and $A_1 \cap B = \emptyset$.

Proposition 3.1. *Suppose that κ is an aleph, X is a Hausdorff space, and $A_0, A_1 \subseteq \kappa^{\mathbb{N}} \times X$ are disjoint sets with uniformly analytic sections. Then they are separated by a set with uniformly Borel sections.*

Proof. Fix closed sets $C_i \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_i: C_i \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A_i = \phi_i(C_i)$, and define $A_{i,s,t} = \phi_i(C_i \cap (\mathcal{N}_s \times \mathcal{N}_t))$ for all $i < 2$, $s \in \kappa^{<\mathbb{N}}$, and $t \in \mathbb{N}^{<\mathbb{N}}$.

Following the usual convention, we identify $\bigcup_{n \in \mathbb{N}} \kappa^n \times \mathbb{N}^n \times \mathbb{N}^n$ with $(\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$. Set $T = \{(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}} \mid A_{0,s,t} \cap \overline{A_{1,s,u}} \neq \emptyset\}$.

Lemma 3.2. *The tree T is well-founded.*

Proof. Suppose, towards a contradiction, that $(\sigma, c_0, c_1) \in [T]$. If $i < 2$, then $(\sigma, c_i) \in C_i$, so we can set $x_i = (\text{proj}_X \circ \phi_i)(\sigma, c_i)$. As A_0 and A_1 are disjoint, it follows that $x_0 \neq x_1$, so there are disjoint open neighborhoods U_0 and U_1 of x_0 and x_1 . Fix $n \in \mathbb{N}$ such that $A_{i,\sigma \upharpoonright n, c_i \upharpoonright n} \subseteq \kappa^{\mathbb{N}} \times U_i$ for all $i < 2$. Then $A_{0,\sigma \upharpoonright n, c_0 \upharpoonright n} \cap \overline{A_{1,\sigma \upharpoonright n, c_1 \upharpoonright n}} = \emptyset$, which contradicts the fact that $(\sigma \upharpoonright n, c_0 \upharpoonright n, c_1 \upharpoonright n) \in T$. $\square$

Recursively define $B_{s,t,u} \subseteq \kappa^{\mathbb{N}} \times X$ for $(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$ by

$$B_{s,t,u} = \begin{cases} \bigcap_{r \in S_s} \bigcup_{m \in \mathbb{N}} \bigcap_{n \in \mathbb{N}} B_{r, t \smallfrown (m), u \smallfrown (n)} & \text{if } (s, t, u) \in T \text{ and} \\ \sim A_{1,s,u} & \text{otherwise,} \end{cases}$$

where $S_s = \{s \smallfrown (\alpha) \mid \alpha < \kappa\}$. The obvious induction shows that $B_{s,t,u}$ has uniformly Borel sections and separates $A_{0,s,t}$ from $A_{1,s,u}$, so $B_{\emptyset, \emptyset, \emptyset}$ separates A_0 from A_1 . $\square$

Given sets $A_i \subseteq \kappa^{\mathbb{N}} \times X_i$ for all $i < 2$ and $R \subseteq \kappa^{\mathbb{N}} \times (X_0 \times X_1)$, we say that (A_0, A_1) is R -independent if $\forall \sigma \in \kappa^{\mathbb{N}} ((A_0)_\sigma \times (A_1)_\sigma) \cap R_\sigma = \emptyset$.

Proposition 3.3. *Suppose that κ is an aleph, X_0, X_1 are Hausdorff spaces, the sets $A_0 \subseteq \kappa^{\mathbb{N}} \times X_0$, $A_1 \subseteq \kappa^{\mathbb{N}} \times X_1$, and $R \subseteq \kappa^{\mathbb{N}} \times (X_0 \times X_1)$ have uniformly analytic sections, and (A_0, A_1) is R -independent. Then there are sets $B_0 \subseteq \kappa^{\mathbb{N}} \times X_0$ and $B_1 \subseteq \kappa^{\mathbb{N}} \times X_1$ with uniformly Borel sections such that $A_0 \subseteq B_0$, $A_1 \subseteq B_1$, and (B_0, B_1) is R -independent.*

Proof. The set $A'_0 = \text{proj}_{\kappa^{\mathbb{N}} \times X_0}(R \cap \text{proj}_{\kappa^{\mathbb{N}} \times X_1}^{-1}(A_1))$ has uniformly analytic sections by two applications of Proposition 1.2. As $A_0 \cap A'_0 = \emptyset$,

Proposition 3.1 yields a set $B_0 \subseteq \kappa^{\mathbb{N}} \times X_0$ with uniformly Borel sections separating A_0 from A'_0 .

Along similar lines, the set $A'_1 = \text{proj}_{\kappa^{\mathbb{N}} \times X_1}(R \cap \text{proj}_{\kappa^{\mathbb{N}} \times X_0}^{-1}(B_0))$ has uniformly analytic sections by Propositions 1.2, 1.5, and 1.6. As $A_1 \cap A'_1 = \emptyset$, Proposition 3.1 yields a set $B_1 \subseteq \kappa^{\mathbb{N}} \times X_1$ with uniformly Borel sections separating A_1 from A'_1 . $\square$

Given a set $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$, we say that A is G -independent if (A, A) is G -independent.

Proposition 3.4. *Suppose that κ is an aleph, X is a Hausdorff space, the sets $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ have uniformly analytic sections, and A is G -independent. Then there exists $B \subseteq \kappa^{\mathbb{N}} \times X$ with uniformly Borel sections such that $A \subseteq B$ and B is G -independent.*

Proof. By Proposition 3.3, there exist $B_0, B_1 \subseteq \kappa^{\mathbb{N}} \times X$ with uniformly Borel sections such that $A \subseteq B_0 \cap B_1$ and (B_0, B_1) is G -independent. Then the set $B_0 \cap B_1$ is as desired. $\square$

4. THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

We use the notation and basic facts concerning $\mathbb{G}_0$, colorings, and homomorphisms from [dRM21, §1].

Proof of Theorem 1. Define $p_i: \kappa^{\mathbb{N}} \times (X \times X) \rightarrow \kappa^{\mathbb{N}} \times X$ by setting $p_i(\sigma, (x_0, x_1)) = (\sigma, x_i)$ for all $i < 2$, $\sigma \in \kappa^{\mathbb{N}}$, and $x_0, x_1 \in X$. As Propositions 1.1 and 1.2 ensure that $\bigcup_{i < 2} p_i(G)$ has uniformly analytic sections, there are closed sets $C, D \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ and $\psi: D \rightarrow \kappa^{\mathbb{N}} \times (X \times X)$ such that $\bigcup_{i < 2} p_i(G) = \phi(C)$ and $G = \psi(D)$.

An *approximation* is a triple $a = (n^a, \phi^a, (\psi_k^a)_{k < n^a})$, where $n^a \in \mathbb{N}$, $\phi^a: 2^{n^a} \rightarrow \mathbb{N}^{n^a}$, and $\psi_k^a: 2^{n^a - (k+1)} \rightarrow \mathbb{N}^{n^a}$ for all $k < n^a$. An approximation b is a *one-step extension* of a if:

- (1) $n^b = n^a + 1$,
- (2) $\forall u \in 2^{n^a} \forall v \in 2^{n^b} (u \sqsubseteq v \implies \phi^a(u) \sqsubseteq \phi^b(v))$, and
- (3) $\forall k < n^a \forall u \in 2^{n^a - (k+1)} \forall v \in 2^{n^b - (k+1)} (u \sqsubseteq v \implies \psi_k^a(u) \sqsubseteq \psi_k^b(v))$.

Given $\sigma \in \kappa^{\mathbb{N}}$, a σ -*configuration* is a triple $\gamma = (n^\gamma, \phi^\gamma, (\psi_k^\gamma)_{k < n^\gamma})$, where $\phi^\gamma: 2^{n^\gamma} \rightarrow \mathbb{N}^{n^\gamma}$ and $\psi_k^\gamma: 2^{n^\gamma - (k+1)} \rightarrow \mathbb{N}^{n^\gamma}$ for all $k < n^\gamma$, such that:

- (1) $\forall u \in 2^{n^\gamma} \phi^\gamma(u) \in C_\sigma$,
- (2) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)} \psi_k^\gamma(v) \in D_\sigma$, and
- (3) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)}$

$$(\text{proj}_{X \times X} \circ \psi_\sigma \circ \psi_k^\gamma)(v) = ((\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_k \frown (i) \frown v))_{i < 2}.$$

We say that γ is *compatible* with $Y \subseteq X$ if $(\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(2^{n^\gamma}) \subseteq Y$, and *compatible* with a if:

- (1) $n^\gamma = n^a$,
- (2) $\forall u \in 2^{n^a} \phi^a(u) \sqsubseteq \phi^\gamma(u)$, and
- (3) $\forall k < n^a \forall v \in 2^{n^a - (k+1)} \psi_k^a(v) \sqsubseteq \psi_k^\gamma(v)$.

Given a set $Z \subseteq \kappa^\mathbb{N} \times X$, we say that a is Z -terminal at σ if no σ -configuration is compatible with Z_σ and a one-step extension of a . Let $A(a, Z)$ denote the set of pairs of the form $(\sigma, (\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_{n^a}))$, where $\sigma \in \kappa^\mathbb{N}$ and γ is a σ -configuration compatible with a and Z_σ .

Lemma 4.1. *Suppose that a is an approximation, $\sigma \in \kappa^\mathbb{N}$, $Z \subseteq \kappa^\mathbb{N} \times X$, and a is Z -terminal at σ . Then $A(a, Z)_\sigma$ is G_σ -independent.*

Proof. See, for example, the proof of [dRM21, Lemma 1.5]. $\square$

Lemma 4.2. *If a is an approximation and $B \subseteq \kappa^\mathbb{N} \times X$ has uniformly Borel sections, then $A(a, B)$ has uniformly analytic sections.*

Proof. The set of pairs (σ, γ) for which γ is a σ -configuration compatible with a is closed, and therefore has uniformly Borel sections by Proposition 1.4, so the set of such pairs for which γ is also compatible with B_σ has uniformly Borel sections by Proposition 1.5, and therefore uniformly analytic sections by Proposition 1.6. The desired result therefore follows from Proposition 1.2. $\square$

Suppose that $U \subseteq \kappa^\mathbb{N}$. We say that a is Z -terminal on U if it is Z -terminal at σ for all $\sigma \in U$. Set $A_s(a, Z) = A(a, Z) \cap (\mathcal{N}_s \times X)$. If $B \subseteq \kappa^\mathbb{N} \times X$ has uniformly Borel sections and a is B -terminal on $\mathcal{N}_s$, then Proposition 3.4 and Lemmas 4.1 and 4.2 yield a G -independent set $A_s(a, B) \subseteq B_s(a, B) \subseteq \mathcal{N}_s \times X$ with uniformly Borel sections.

Let $\mathcal{A}$ denote the set of all approximations. We will recursively construct a decreasing sequence $(B^\alpha)_{\alpha < \kappa^+}$ of subsets of $\kappa^\mathbb{N} \times X$ with uniformly Borel sections and an increasing sequence $(S^\alpha)_{\alpha < \kappa^+}$ of subsets of $\kappa^{<\mathbb{N}} \times \mathcal{A}$, starting with $S^0 = \emptyset$ and setting $S^\delta = \bigcup_{\alpha < \delta} S^\alpha$ for all limit ordinals $\delta < \kappa^+$. Suppose that $\alpha < \kappa^+$ and we have already constructed $(B^\beta)_{\beta < \alpha}$ and $(S^\beta)_{\beta \leq \alpha}$. Define $\rho^\alpha: S^\alpha \rightarrow \alpha$ by $\rho^\alpha(s, a) = \min\{\beta < \alpha \mid (s, a) \in S^{\beta+1}\}$, set $B^\alpha = \sim \nabla_{s \in \kappa^{<\mathbb{N}}} \bigcup_{(s, a) \in S^\alpha} B_s(a, B^{\rho^\alpha(s, a)})$ and $S^{\alpha+1} = S^\alpha \cup \{(s, a) \in \sim S^\alpha \mid a \text{ is } B^\alpha\text{-terminal on } \mathcal{N}_s\}$, and fix $B_s(a, B^\alpha)$ as above for all $(s, a) \in S^{\alpha+1} \setminus S^\alpha$. As $|\kappa^{<\mathbb{N}} \times \mathcal{A}| = \kappa$, there is a least $\lambda < \kappa^+$ with $S^\lambda = S^{\lambda+1}$. Set $B = B^\lambda$, $S = S^\lambda$, and $\rho = \rho^\lambda$.

Lemma 4.3. *Suppose that $\alpha < \kappa^+$ and a is an approximation that is not $B^{\alpha+1}$ -terminal on $\mathcal{N}_s$ for some $s \in \kappa^{n^a}$. Then there is a one-step extension b of a that is not B^α -terminal on $\mathcal{N}_{s \smallfrown (\xi)}$ for some $\xi < \kappa$.*

Proof. Fix $\sigma \in \mathcal{N}_s$ for which a is not $B^{\alpha+1}$ -terminal at σ . Then there is a σ -configuration γ compatible with a one-step extension b

of a and $B_\sigma^{\alpha+1}$. Set $\xi = \sigma(n^a)$. If b is B^α -terminal on $\mathcal{N}_{s \smallfrown (\xi)}$ and $\beta = \rho^{\alpha+1}(s \smallfrown (\xi), b)$, then $(\sigma, (\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(\mathfrak{s}_{nb})) \in A_{s \smallfrown (\xi)}(b, B^\beta) \cap B^{\alpha+1} \subseteq B_{s \smallfrown (\xi)}(b, B^\beta) \cap B^{\alpha+1} = \emptyset$, the desired contradiction. $\square$

Let a_0 denote the unique approximation for which $n^{a_0} = 0$.

Lemma 4.4. *Suppose that a_0 is not B -terminal on $\kappa^\mathbb{N}$. Then there is a continuous homomorphism from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^\mathbb{N}$.*

Proof. By Lemma 4.3 there exist $a_n \in \mathcal{A}$ with $n^{a_n} = n$, $s_n \in \kappa^n$, $\sigma_n \in \mathcal{N}_{s_n}$, and a σ_n -configuration γ_n compatible with a_{n+1} and B_{σ_n} , where a_{n+1} is a one-step extension of a_n and $s_n \sqsubseteq s_{n+1}$, for all $n \in \mathbb{N}$.

Set $\sigma = \bigcup_{n \in \mathbb{N}} s_n$. Then $\sigma_n \rightarrow \sigma$. Define $\phi': 2^\mathbb{N} \rightarrow \mathbb{N}^\mathbb{N}$ and $\psi'_k: 2^\mathbb{N} \rightarrow \mathbb{N}^\mathbb{N}$ by $\phi'(c) = \bigcup_{n \in \mathbb{N}} \phi^{a_n}(c \upharpoonright n)$ and $\psi'_k(c) = \bigcup_{n > k} \psi_k^{a_n}(c \upharpoonright (n - (k+1)))$. If $c \in 2^\mathbb{N}$ and $k \in \mathbb{N}$, then $(\sigma_n, \phi^{\gamma_n}(c \upharpoonright n+1)) \rightarrow (\sigma, \phi'(c))$ and $(\sigma_n, \psi_k^{\gamma_n}(c \upharpoonright (n-k))) \rightarrow (\sigma, \psi'_k(c))$, so the fact that C and D are closed ensures that $(\sigma, \phi'(c)) \in C$ and $(\sigma, \psi'_k(c)) \in D$.

Define $\pi: 2^\mathbb{N} \rightarrow X$ by $\pi(c) = (\text{proj}_X \circ \phi_\sigma)(\phi'(c))$ and observe that $(\text{proj}_{X \times X} \circ \psi_\sigma)(\psi'_k(c)) = (\pi(\mathfrak{s}_k \smallfrown (i) \smallfrown c))_{i < 2}$ for all $k \in \mathbb{N}$ and $c \in 2^\mathbb{N}$, so π is a continuous homomorphism from $\mathbb{G}_0$ to G_σ . $\square$

By Lemma 4.4, we can assume that a_0 is B -terminal on $\kappa^\mathbb{N}$. Set $\alpha = \rho(\emptyset, a_0)$. Then $A(a_0, B) = B \cap \bigcup_{i < 2} p_i(G)$ and $A(a_0, B^\alpha) \subseteq B_\emptyset(a_0, B^\alpha)$, so $A(a_0, B) = \emptyset$, thus B is G -independent. For all $a \in \mathcal{A}$ and $n \in \mathbb{N}$, define $S_{a,n} = \{s \in \kappa^n \mid (s, a) \in S\}$ and $B_{a,n} = \nabla_{s \in S_{a,n}} B_s(a, B^{\rho(s,a)})$. Then the latter sets are also G -independent, so (1) follows from the fact that $\mathcal{A}$ is countable and $\kappa^\mathbb{N} \times X = B \cup \bigcup_{a \in \mathcal{A}, n \in \mathbb{N}} B_{a,n}$.

It only remains to note that if both (1) and (2) hold, then there is a Borel $\mathbb{N}$ -coloring of $\mathbb{G}_0$, contradicting [dRM21, Proposition 1.3]. $\square$

5. STRATIFICATIONS AND CATEGORY TRANSFORMS

For all sets $B \subseteq \kappa^\mathbb{N} \times X$ and sequences $t \in \kappa^{<\mathbb{N}}$, define

$$B^{*t} = \{x \in X \mid B^x \text{ is comeager in } \mathcal{N}_t\}.$$

Proposition 5.1. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, $B \subseteq \kappa^\mathbb{N} \times X$ and $B_s \subseteq \kappa^\mathbb{N} \times X$ have uniformly Borel sections for all $s \in \kappa^{<\mathbb{N}}$, and $U \subseteq X$. Then:*

- (1) $(\kappa^\mathbb{N} \times U)^{*t} = U$.
- (2) $(\sim B)^{*t} = \sim \bigcup_{u \sqsupseteq t} B^{*u}$.
- (3) $(\nabla_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t} = \bigcap_{u \sqsupseteq t} \bigcup_{v \sqsupseteq u} \bigcup_{n \leq |v|} B_{v|n}^{*v}$.

Proof. It is clear that (1) holds, and Proposition 2.1 reduces (2) to localization. To see (3), note that if $x \in (\nabla_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t}$, then $(\nabla_{s \in \kappa^{<\mathbb{N}}} B_s)^x$

is comeager in $\mathcal{N}_t$, so $\forall u \supseteq t \exists n \in \mathbb{N} \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap B_s^x$ is non-meager in $\mathcal{N}_u$, thus $\forall u \supseteq t \exists s \in \kappa^{<\mathbb{N}} \mathcal{N}_s \cap B_s^x$ is non-meager in $\mathcal{N}_u$, and it follows that $\forall u \supseteq t \exists v \supseteq u \exists n \leq |v| B_{v \upharpoonright n}^x$ is comeager in $\mathcal{N}_v$, hence $x \in \bigcap_{u \supseteq t} \bigcup_{v \supseteq u} \bigcup_{n \leq |v|} B_{v \upharpoonright n}^{*v}$. And if $x \notin (\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t}$, then $(\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is not comeager in $\mathcal{N}_t$, so there exists $u \supseteq t$ such that $(\bigcap_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is meager in $\mathcal{N}_u$, thus $\forall v \supseteq u \forall n \leq |v| B_{v \upharpoonright n}^x$ is meager in $\mathcal{N}_v$, hence $x \notin \bigcup_{v \supseteq u} \bigcup_{n \leq |v|} B_{v \upharpoonright n}^{*v}$. $\square$

A subset of a topological space is κ^+ -Borel if it belongs to the smallest family containing the open sets and closed under complements and unions of cardinality at most κ .

Proposition 5.2. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then B^{*t} is κ^+ -Borel.*

Proof. By the obvious induction using Proposition 5.1. $\square$

A set $A \subseteq X$ is κ -Souslin if there exist a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$. We say that a set $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a *stratification* of a κ -Souslin set A if there exist a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ such that $A = \phi(C)$ and $A' = \phi'(C')$, where $C' = \{(\sigma, b) \in \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \mid \sigma \circ b \in C\}$ and $\phi': C' \rightarrow \kappa^{\mathbb{N}} \times X$ is given by $\phi'(\sigma, b) = (\sigma, \phi(\sigma \circ b))$. For each $\sigma \in \kappa^{\mathbb{N}}$, we view A'_σ as an *analytic trace* of A on $\sigma(\mathbb{N})$.

Proposition 5.3. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq X$ is κ -Souslin, $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a stratification of A , and $x \in A$. Then $\forall \sigma \in \kappa^{\mathbb{N}} x \in A'_\sigma$.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$ and $A' = \phi'(C')$, as well as $\tau \in C$ for which $\phi(\tau) = x$. Then the set $H = \bigcap_{n \in \mathbb{N}} \bigcup_{s \in \kappa^{<\mathbb{N}}} \mathcal{N}_{s \frown (\tau \upharpoonright n)}$ is a dense G_δ subset of $\kappa^{\mathbb{N}}$ and $\forall \sigma \in H x \in A'_\sigma$. $\square$

Proposition 5.4. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a Hausdorff space, G is a κ -Souslin digraph on X , G' is a stratification of G , and $B \subseteq \kappa^{\mathbb{N}} \times X$ is a G' -independent set with uniformly Borel sections. Then B^{*t} is G -independent.*

Proof. Otherwise fix $x, y \in B^{*t}$ with $x G y$ and appeal to Proposition 5.3 to obtain $\sigma \in \mathcal{N}_t$ with $x, y \in B_\sigma$ and $x G'_\sigma y$, a contradiction. $\square$

Theorem 5.5 (Kanovei). *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and G is a κ -Souslin digraph on X . Then at least one of the following holds:*

- (1) *There is a κ^+ -Borel coloring $c: X \rightarrow \kappa$ of G .*
- (2) *There is a continuous homomorphism from $\mathbb{G}_0$ to G .*

Proof. Fix a stratification G' of G . Then $\forall \sigma \in \kappa^{\mathbb{N}} G'_\sigma \subseteq G$, so Theorem 1 allows us to assume there is a function $c': \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$, whose fibers have uniformly Borel sections, such that $\forall \sigma \in \kappa^{\mathbb{N}} c'_\sigma$ is a coloring of G'_σ . For all $n \in \mathbb{N}$, set $B_n = (c')^{-1}(\{n\})$. For all $t \in \kappa^{<\mathbb{N}}$, Proposition 5.2 ensures that B_n^{*t} is κ^+ -Borel and Proposition 5.4 ensures that it is G -independent. But if $x \in X$, then $\kappa^{\mathbb{N}} = \bigcup_{n \in \mathbb{N}} B_n^x$, so there exists $n \in \mathbb{N}$ for which B_n^x is non-meager. Proposition 2.1 and localization therefore yield $t \in \kappa^{<\mathbb{N}}$ for which B_n^x is comeager in $\mathcal{N}_t$, thus $x \in B_n^{*t}$. $\square$

As noted in [Mil12, §8], the weakening of Theorem 5.5 without the definability condition on c follows from the simplification of the classical proof of the $\mathbb{G}_0$ dichotomy in which the use of separation is removed. As separation easily generalizes to κ -Souslin sets, this suggests a simpler route towards a classical proof of the full version of Theorem 5.5. The direct adaptation of that proof, however, would require separation for continuous images of κ^+ -Borel sets. To see that this is consistently false, note first that every Π_2^1 set is $\aleph_2$ -Borel (see, for example, [Kec95, Exercise 38.8]), so every Σ_3^1 set is a continuous image of an $\aleph_2$ -Borel set. But if $\mathbf{MA} + \neg\mathbf{CH}$ holds and $\omega_1 = \omega_1^L$, then every $\aleph_2$ -Borel set has the Baire property but there is a Δ_3^1 set B without the Baire property by [HS85, Claim 8], so B and $\sim B$ are not separable by an $\aleph_2$ -Borel set.

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